

COURSES STUDIED & TOPICS

Personal Dataset — Ikram Wazir

Programming / AI

- Programming for AI
- Python
- Artificial Intelligence

Machine Learning

- Machine Learning
- Linear Regression
- Naive Bayes
- Decision Trees
- TF-IDF
- LinearSVC
- Label Encoding
- Cosine Similarity
- Confusion Matrix

Deep Learning

- ANN
- RNN
- LSTM
- Bi-LSTM
- LSTM gates, hidden state and cell state
- Temporal Fusion Transformer

NLP

- Preprocessing
- Tokenization
- Stopwords
- Negation
- Stemming
- Porter Stemmer
- WordNet Lemmatizer
- TF-IDF/IDF
- Statistical Language Models
- Unigram
- Information Retrieval
- Text Similarity

- Generative AI
- RAG
- LangChain concepts

Data / Systems

- Data Warehousing
- Data Engineering
- FastF1
- TimeSHAP

Mathematics

- Linear Algebra
- Vector spaces and axioms
- Linear combinations
- Span
- Linear independence
- Basis
- Coordinates

Communication

- EN-220 Communication Skills
- Resume
- Cover Letter
- Email
- Business Letter
- Questionnaire
- Abstract
- Report
- Reflection

Historical Planned Areas

- Computer Vision planned for 7th semester; Generative AI planned for 8th semester; NLP excluded from a personal summer roadmap because it was being studied as a university summer course.